Research Methods in Computer Science

CHAPTER SIX:
DATA COLLECTION AND ANALYSIS

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- Data Collection (primary and secondary data)
- Analysis of Data with case studies

Data Collection (primary and secondary data)

- **Data Collection** refers to the systematic process of **gathering, measuring**, and **analysing information** from various sources to get a complete and accurate picture of an area of interest.
- Different sources of data collection include
  - **Primary Sources** and
  - **Secondary Sources**.

What is Data?

- Data is a **collection of measurements** and **facts** and a tool that helps an individual or a group of individuals reach a sound conclusion by providing them with some information.
- It helps the analyst understand, analyze, and interpret different socio-economic problems like unemployment, poverty, inflation, etc.

Data Collection (primary and secondary data)

Sources of Collection of Data

- **Primary Source**
  - It is a collection of data from the **source of origin.** It provides the researcher with **first-hand quantitative** and **raw information** related to the statistical study.
  - The data collected by the investigator from **primary sources for the first time from scratch** is known as **primary data.**
  - This data is collected directly from the source of origin. It is real-time data and is always specific to the researcher's needs. The primary data is available in raw form.
  - the **accuracy** and **reliability** of primary data are **more than** the **secondary data.**
  - the primary sources of data give the researcher direct access to the subject of research.
    - **E.g. statistical data, works of art, and interview transcripts**

Data Collection (primary and secondary data)

Sources of Collection of Data

- **Secondary Source**
  - It is a **collection of data from some institutions or agencies** that have already collected the data through primary sources.
  - The data already in existence which has been previously collected by someone else for other purposes is known as secondary data.
  - It **does not include any real-time data** as the research has already been done on that information. However, the cost of collecting secondary data is less. As the data has already been collected in the past, it can be found in refined form.
  - The chances of finding the exact information or data specific to the researcher's needs are less. However, the time required to collect secondary data is short and hence is a quick and easy process.
  - The **accuracy** and **reliability** of **secondary data** are relatively **less than** the **primary data.**
    - **E.g. Reviews, government websites containing surveys, academic books, published journals, articles, etc.**

Data Collection (primary and secondary data)

Methods of Collecting Primary Data

- Direct Personal Investigation
- Indirect Oral Investigation
- Information from Local Sources or Correspondents
- Information through Questionnaires and Schedules

Data Collection (primary and secondary data)

Methods of Collecting Secondary Data

- **Published Sources**
- Published sources refer to materials that have undergone the process of publication. **Examples** include,
  - Government Publications, Semi-Government Publications, Publications of Trade Associations, Journals and Papers, International Publications, Publications of Research Institutions
- **Unpublished Sources**
  - The data in unpublished sources is collected by different government organizations and other organizations. These organizations usually collect data for their self-use and are not published anywhere.
  - **E.g.** research work done by professors, professionals, teachers and records maintained by business and private enterprises.

Analysis of Data with case studies

- **Case study research** involves a detailed and intensive analysis of a particular event, situation, organization, or social unit.
- A **case** has a defined **space** and **time** frame: "a phenomenon of some sort in a bounded context" (*Miles, Huberman, & Saldaña*)
- In terms of scope, a **case study** is an in-depth investigation of a contemporary phenomenon within its real-life context.
- A case could be an individual, a role, a small group, an organization, a community, or even a nation (Miles et al.)
- Case studies are best conceptualized by the bounded phenomena of interest and not by specific methods; thus, different studies can be conducted under this broad umbrella.

Analysis of Data with case studies

- **Analysis of data with case studies** refers to a **research methodology** that involves **gathering** and **analysing** comprehensive information about certain people, groups, or circumstances (case studies) in order to obtain a deeper understanding of a phenomenon.
- Various data analysis techniques are frequently used to find patterns, trends, and linkages within the case data.

Benefits of Using a Case Study Approach

- help you focus your research within the confines of space and time on a specific case.
- gives you an opportunity to collect different kinds of data, such as interviews, documents, observations, surveys, and others, about the case
- provides you with the chance to get an in-depth look at an organization or individual and the inner workings and interactions of that organization or individual.

Analysis of Data with case studies

Well-known case studies:

- **AlphaGo vs. Lee Sedol:** The DeepMind AI program AlphaGo defeating professional Go player Lee Sedol, showcasing the power of deep reinforcement learning.
- **ImageNet Challenge:** A benchmark competition where AI models are tested on image classification tasks, demonstrating the progress in computer vision.
- **MNIST Handwritten Digits Dataset:** A widely used dataset for testing machine learning algorithms on basic digit recognition. (**you can use it as a case to analyze model performance**)
- **Spam Email Filtering:** Using machine learning models to classify emails as spam or not, demonstrating practical applications of classification algorithms.
- **The ARPANET:** The precursor to the modern internet, demonstrating the concept of packet switching.

Analysis of Data with case studies

Conducting Case Study Research

- The first research thoughts begin with a research problem, which involves the identification of a lack of knowledge about some issue or phenomenon.
- Identification of the problem then leads to crafting the purpose statement and research question(s)
- Examining the research question(s) then allows you to determine whether a case study is the appropriate design.

Case Study Sample Size

- Typically, a case study has a sample of one (i.e., the bounded case, but note that sampling can also occur within the case), unless the research project is a multiple-case study. In a multiple-case study, having three to four distinct cases for comparison is probably the most cases that one can realistically handle

Analysis of Data with case studies

Conducting Case Study Research

Case Study Sample Selection

- Sample sizes in case studies are typically small, which is common in most qualitative research.
- Sometimes the selection of samples and cases to use is straightforward and clear, due to the uniqueness of the person or organization or because of special arrangements or access to the case.
- In some situations, however, there may be many qualified case study candidates, and you may have to use a screening procedure to select the proper ones.

Analysis of Data with case studies

Case Study Research Questions

- Research questions for a case study can be both quantitative and qualitative but frequently use terms similar to other qualitative research designs.
- Typically, case study research questions use words such as **how or why.**
- Over- all, the case study research questions need to address the substance of **what (case) the study is about**

More on Analysis of Data with case studies

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